

AI-900 Exam Topics Summary

1. AI & Machine Learning Fundamentals

- Difference between AI, Machine Learning, and Deep Learning
- Labeled vs. unlabeled data
- Features vs. Labels
- Machine Learning lifecycle: data preparation, training, evaluation, deployment

2. Types of Machine Learning Problems

- Classification: binary and multi-class problems (Yes/No, sentiment, disease detection)
- Regression: numeric prediction (prices, sales, sea level)
- Clustering: grouping similar data without labels
- Anomaly Detection: identifying unusual patterns or fraud

3. Model Evaluation & Metrics

- Confusion Matrix (TP, FP, FN, TN)
- Accuracy and its limitations
- Precision, Recall, ROC, AUC
- Regression metrics: RMSE, R^2
- Train, validation, and test data splits

4. Feature Engineering & Selection

- Feature engineering: creating or transforming features
- Feature selection: choosing the most relevant features
- Featurization in Automated ML

5. Azure Machine Learning

- Azure ML Designer: datasets, modules, pipelines
- Data split, model training, and evaluation
- Publishing pipelines and real-time endpoints (AKS)
- Automated ML concepts and explainable models

6. Azure Cognitive Services

- Computer Vision: image classification, object detection, OCR, segmentation
- Face Service: detection, verification, identification, similarity
- Natural Language Processing: sentiment analysis, key phrase extraction, chatbots
- Form Recognizer: extracting text, key-value pairs, and tables
- Custom Vision for custom image models

7. Conversational AI

- Chatbots and automated customer support

- Reducing workload of human agents

8. Responsible AI Principles

- Fairness
- Inclusiveness
- Transparency
- Privacy and Security
- Reliability and Safety
- Accountability
- Explainability of AI decisions

9. Business & Scenario-Based Questions

- Choosing the right ML approach for a given business problem
- Selecting the correct Azure service based on scenarios